Bahria University Lahore Campus

Department of Computer Science

Course Code/number	**CSL-210**
Course Title/Name	**Object Oriented Programming Lab**
Credit Hours/Contact hours	1
Degree Program	BS(CS)
Prerequisites or Corequisites	CSC-113: Computer Programming
Assessment Methods and Weightage	Lab Journals 10 % Lab Evaluation 20 % Lab Midterm 20 % Lab Final Term 30 % Lab Project 20 % **Total** **100%**
Textbook(s)	• H. M. Deitel "C++ How to Program", 5th Ed., Prentice Hall, 2005. Object-Oriented Programming in C++, 4th Ed, Robert Lafore • Deitel and Deitel, Java How to Program, 9th Edition, Prentice Hall, 2012
Reference Material	Lab Manual
Web Resources	-
Instructor Name/Cluster Head/ Subject Expert	**Instructor Name:** Mr. Mughees Ahmad **Status**: Regular **Designation:** Senior Lecturer **Course Instructor Name:** Ms. Fatima Zulfiqar **Cluster Head Name:** Ms. Iram Noureen
Course Aims	The aim of the course is to understand and implement basic to advance level computer programming concepts.

Course Objectives	1. Practical understanding of Abstract Data Types (classes) to build small units to pack the data inside the code 2. How to create classes and objects in C++ 3. Difference between static and dynamic polymorphism and C++ syntax to achieve them 4. Several types of inheritance: linear, multiple, virtual (to resolve diamond shaped hierarchy) 5. Creating generic (type independent) functions and classes 6. Other C++ specific stuff e.g. I/O streams, exceptions handling

Course Outcomes	After completing this course, the student must demonstrate the knowledge and ability to: 1. Compare the object vs the procedural approach to writing software 2. Define and use classes in a C++ program 3. Explain how object-oriented software engineering enhances the software development process 4. Identify the major elements in an object-oriented programming language 5. Implement the concepts of data abstraction and encapsulation in the creation of abstract data types 6. Implement operator overloading 7. Use inheritance in C++ 8. Derive classes using inheritance in C++ 9. Select the proper class protection mechanism 10. Demonstrate the use of virtual functions to implement polymorphism 11. Write programs utilizing the I/O classes in C++ 12. Create and use abstract data types 13. Understand some advanced features of C++ including templates, exceptions, and multiple inheritance 14. Use correct object-oriented terminology
Course Catalogue	-

Lecture Plan (16 Weeks)	**Week #**	**Lecture /Hour #**	**Topic to be covered**	**Learning outcomes**	**Reference Text**
	Week #.1	1 Lab. / 3 Hrs.	**Revision of Programming Concepts**	Revision of Structures and pointers (Pass by value and by reference)	Lab 01, Lab Manual

Week #.2	1 Lab. / 3 Hrs.	**Classes & Objects**	Understanding of Classes & Objects. Data members, Member Functions of Classes	Lab 02, Lab Manual
Week #.3	1 Lab. / 3 Hrs.	**Constructors**	Understanding of Constructors, Non- parameterized Constructors, parameterized Constructors and Constructor Overloading	Lab 03, Lab Manual
Week #.4	1 Lab. / 3 Hrs.	**Array of Objects**	Array of Objects, passing objects as function arguments, Returning objects from functions	Lab 04, Lab Manual
Week #.5	1 Lab. / 3 Hrs.	**Static data Members, Const Member Functions**	Static class data members, const member functions, pass objects and objects array as argument.	Lab 05, Lab Manual
Week #.6	1 Lab. / 3 Hrs.	**Unary Operator Overloading**	Learning of Prefix Operators, Postfix Operators, Operator Returning Value and Nameless Temporary Object	Lab 06, Lab Manual
Week #.7	1 Lab. / 3 Hrs.	**Binary Operator Overloading**	Implementation of Binary Operators: Addition, Subtraction, Multiplication, Division, Modulus operators	Lab 07, Lab Manual

Week #.8	1 Lab. / 3 Hrs.	**MID TERM LAB**		
Week #.9	**M I D T E R M COURSE**			
Week #.10	1 Lab. / 3 Hrs.	**Inheritance**	Implementation of Basic Inheritance	Lab 08, Lab Manual
Week #.11	1 Lab. / 3 Hrs.	**Multiple Inheritance**	Understanding of Multiple File Program, Multiple – Inheritance	Lab 09, Lab Manual
Week #.12	1 Lab. / 3 Hrs.	**Polymorphism**	Implementation of overloading and overriding functions. Understanding of virtual and Pure Virtual functions	Lab 10, Lab Manual
Week #.13	1 Lab. / 3 Hrs.	**Composition and Aggregation**	Implementation of relation between classes using composition and aggregation relation	Lab 11, Lab Manual

	Week #.14	1 Lab. / 3 Hrs.	**Exception Handling**	Concept of Exceptions, exceptions with parameters and understanding of Stacks.	Lab 12, Lab Manual
	Week #.15	1 Lab. / 3 Hrs.	**Templates**	Understanding of Function Templates, Class Templates	Lab 13, Lab Manual
	Week #.16	1 Lab. / 3 Hrs.	**File Handling**	Understanding of File Handling, Reading and Writing Data to a File and File Handling with objects.	Lab 14, Lab Manual
	Final Term				

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/Quiz
	1					
	2					
	3					
	4					
	5					
	6					
	7			Project Assigned		
	8					
	10					
	11					
	12					
	13					
	14					
	15					
	16					Projects Evaluation
	17					Projects Result

Instructor Name: **Mr. Mughees Ahmad**

Instructor Signature: _____

Date: **21/02/2023**